

Deformation Theory of Program Refactoring

Halley Young
Microsoft Research
halleyyoung@microsoft.com

March 22, 2026

Abstract

We develop a deformation-theoretic account of program refactoring. A refactoring of a program P_0 is modelled as a one-parameter deformation P_t over the semantic site $\mathcal{S}$: a flat family of sheaves on $\mathcal{S}$ whose fibre at $t = 0$ is the original program. First-order deformations — the “infinitesimal” refactorings — are classified by the first Čech cohomology group $H^1(\mathcal{S}, T_{\mathcal{S}})$, where $T_{\mathcal{S}}$ is the tangent sheaf of the site. The obstruction to extending a first-order refactoring to a complete one-parameter family lives in $H^2(\mathcal{S}, T_{\mathcal{S}})$ and is computed by the *Kodaira-Spencer map* κ . Our main theorem (Theorem ??) states that a first-order refactoring $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ extends to a full family if and only if $\kappa([\delta]) = 0$ in $H^2(\mathcal{S}, T_{\mathcal{S}})$. When $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$ — we call the program *unobstructed* — every local refactoring extends globally. We give an algorithm that computes the Kodaira-Spencer obstruction before applying a refactoring and uses it as a *safety certificate*: if $\kappa = 0$, the refactoring is guaranteed to preserve verification status. Formal proofs of all algebraic lemmas and the main theorem are given in the companion LEAN 4 development (Paper 22, `Paper22.DeformationTheory.lean`).

Contents

1 Introduction

Refactoring is among the most common activities in software engineering. Renaming a module, extracting a method, inlining a helper, changing a parameter order — each of these operations is *semantic-preserving by intention*, yet each can silently break downstream verification conditions if applied naively. Existing verification tools treat refactoring as a black box: one re-runs the full proof pipeline after every change and hopes for the best. This paper asks a more precise question:

Given a program P_0 that is already verified, and a proposed refactoring R , can we certify in advance that R will not break verification — without re-running the proof?

We answer this question using the deformation theory of complex manifolds, transplanted into the sheaf-theoretic programme of Judgment Geometry. The key insight is that a refactoring is not merely an edit to source text: it is a *deformation* of the program, a parameterized family P_t of semantic objects with P_0 as the “base”. Classical results of Kodaira and Spencer [?] characterize when a first-order (infinitesimal) deformation can be “integrated” to a full family, and the obstruction to integration lives in a cohomology group. We adapt this framework to programs via the semantic site introduced in Paper 1 [?].

Contributions

1. **Tangent sheaf $T_{\mathcal{S}}$** (Section ??). We define the tangent sheaf of the semantic site $\mathcal{S}$ as the sheaf of derivations of the structure sheaf $\mathcal{O}_{\mathcal{S}}$, and give a concrete description of $T_{\mathcal{S}}(U)$ for function, module, and branch coordinates.
2. **Classification of first-order refactorings** (Section ??). We prove that isomorphism classes of first-order deformations of P_0 over the dual numbers $k[\varepsilon]/(\varepsilon^2)$ are in natural bijection with $H^1(\mathcal{S}, T_{\mathcal{S}})$.
3. **Kodaira-Spencer obstruction** (Section ??). We construct the Kodaira-Spencer map $\kappa: H^1(\mathcal{S}, T_{\mathcal{S}}) \rightarrow H^2(\mathcal{S}, T_{\mathcal{S}})$ and prove that it is the complete obstruction to extension.
4. **Main theorem** (Section ??). We prove Theorem ??: $[\delta]$ extends to a full one-parameter family if and only if $\kappa([\delta]) = 0$.
5. **Safe refactoring algorithm** (Section ??). We give an algorithm that computes κ on the Čech complex of the program, and use it as a pre-flight check before applying a refactoring.

Organisation

Section ?? reviews the classical deformation theory of complex manifolds; the reader familiar with Kodaira-Spencer theory may skip it. Section ?? recalls the semantic site and its structure sheaves. Section ?? defines $T_{\mathcal{S}}$. Sections ??–?? develop the cohomological machinery. Section ?? states and proves the main theorem. Section ?? gives the safe-refactoring application. Section ?? works through three concrete examples. Section ?? discusses related work and Section ?? concludes.

2 Background: Deformation Theory

We recall the classical theory of Kodaira-Spencer, following Hartshorne [?] and Schlessinger [?]. Throughout, k denotes a field of characteristic zero.

2.1 Deformations of complex manifolds

Let X_0 be a compact complex manifold. A *deformation* of X_0 is a proper flat holomorphic map $\pi: \mathcal{X} \rightarrow B$ with a distinguished point $0 \in B$ such that $\pi^{-1}(0) \cong X_0$. The parameter space B is a small contractible neighbourhood of 0 in $\mathbb{C}$. Intuitively, the fibres $X_t = \pi^{-1}(t)$ are “small perturbations” of X_0 as t moves away from 0.

A *first-order deformation* (or *infinitesimal deformation*) is a deformation over the dual numbers $\text{Spec}(k[\varepsilon]/(\varepsilon^2))$: a flat sheaf on X_0 of $k[\varepsilon]/(\varepsilon^2)$ -algebras lifting $\mathcal{O}_{X_0}$. Such deformations are classified by $H^1(X_0, T_{X_0})$, the first cohomology of the holomorphic tangent sheaf.

2.2 The Kodaira-Spencer map

Given a smooth family $\pi: \mathcal{X} \rightarrow B$, the *Kodaira-Spencer map* is the derivative

$$\rho_0: T_{B,0} \longrightarrow H^1(X_0, T_{X_0}),$$

sending a tangent vector $\partial/\partial t|_{t=0}$ to the cohomology class of the deformation. Explicitly, if $\{U_i\}$ covers X_0 by coordinate charts, the transition functions $f_{ij}(t, z)$ satisfy $f_{ij}(0, z) = f_{ij}^0(z)$ and their t -derivatives give the Čech 1-cocycle $\partial f_{ij}/\partial t|_{t=0}$.

2.3 Obstructions and Schlessinger’s theory

The obstruction to extending a first-order deformation $\xi \in H^1(X_0, T_{X_0})$ to a second-order deformation over $k[t]/(t^3)$ is a class $\text{ob}(\xi) \in H^2(X_0, T_{X_0})$, computed via the bracket $[\xi, \xi]$ in the graded Lie algebra $H^*(X_0, T_{X_0})$.

Theorem 2.1 (Classical Kodaira-Spencer [?]). *A first-order deformation ξ extends to all orders if and only if $\text{ob}(\xi) = 0$ in $H^2(X_0, T_{X_0})$. In particular, when $H^2(X_0, T_{X_0}) = 0$, every deformation extends and the manifold X_0 is called unobstructed.*

Our programme is to transport this theorem from complex manifolds to programs, replacing X_0 with a program P_0 and $H^*(X_0, T_{X_0})$ with $H^*(\mathcal{S}, T_{\mathcal{S}})$.

3 Programs as Geometric Objects

We work with the semantic site $\mathcal{S}$ from Paper 1 [?]. We recall the definitions needed here; for full details see that paper.

3.1 The semantic site

The *semantic site* $\mathcal{S} = (C, J)$ consists of a small category C of program coordinates and a Grothendieck topology J on C . Objects of C are coordinates

$$U \in \{\text{module, function, branch, loop body, expression, } \dots\},$$

and morphisms $U \rightarrow V$ are structural inclusions or transport maps (call site $\rightarrow$ callee, branch $\rightarrow$ parent conditional, $\cdot$). A *covering family* $\text{Cov}(U) = \{U_i \rightarrow U\}_{i \in I}$ is a family of morphisms that jointly cover U : for example, the branches $\{U_{\text{then}}, U_{\text{else}}\}$ cover a conditional U_{if} .

3.2 The structure sheaf $\mathcal{O}_{\mathcal{S}}$

For each coordinate U , the *semantic algebra* $\mathcal{O}_{\mathcal{S}}(U)$ is the ring of program observations at U :

$$\mathcal{O}_{\mathcal{S}}(U) = \{\text{specifications, type signatures, preconditions, postconditions, invariants at } U\}.$$

Restriction maps $\rho_V^U: \mathcal{O}_{\mathcal{S}}(U) \rightarrow \mathcal{O}_{\mathcal{S}}(V)$ for $V \hookrightarrow U$ send a specification to its restriction to the subregion V . The sheaf axiom says that specifications glue: a compatible family of local specifications over a cover $\{U_i\}$ of U glues uniquely to a global specification on U .

3.3 Programs as sheaves

A *program* P_0 on $\mathcal{S}$ is a section of $\mathcal{O}_{\mathcal{S}}$: an assignment of a concrete semantic value $P_0(U) \in \mathcal{O}_{\mathcal{S}}(U)$ to each coordinate U , compatible with restriction. The verification condition for P_0 is the sheaf-descent condition studied in Paper 3 [?]: local sections must glue consistently.

4 The Tangent Sheaf $T_{\mathcal{S}}$

4.1 Kähler differentials

Definition 4.1 (Sheaf of Kähler differentials). The sheaf of *Kähler differentials* $\Omega_{\mathcal{S}}^1$ on $\mathcal{S}$ is the $\mathcal{O}_{\mathcal{S}}$ -module generated by symbols $\{df \mid f \in \mathcal{O}_{\mathcal{S}}(U)\}$ subject to:

1. $d(f + g) = df + dg$,
2. $d(fg) = f dg + g df$ (Leibniz rule),
3. $d(c) = 0$ for every constant $c \in k$.

Intuitively, $\Omega_{\mathcal{S}}^1(U)$ is the “cotangent space” of the program at U : the module of infinitesimal changes to specifications. A generator df records a small variation of the semantic observable f .

4.2 Definition of $T_{\mathcal{S}}$

Definition 4.2 (Tangent sheaf). The *tangent sheaf* of the semantic site is the $\mathcal{O}_{\mathcal{S}}$ -module dual:

$$T_{\mathcal{S}} := \mathcal{H}om_{\mathcal{O}_{\mathcal{S}}}(\Omega_{\mathcal{S}}^1, \mathcal{O}_{\mathcal{S}}).$$

Equivalently, $T_{\mathcal{S}}(U) = \text{Der}(\mathcal{O}_{\mathcal{S}}(U))$, the k -module of *derivations*:

$$T_{\mathcal{S}}(U) = \{D: \mathcal{O}_{\mathcal{S}}(U) \rightarrow \mathcal{O}_{\mathcal{S}}(U) \mid D(fg) = f D(g) + g D(f)\}.$$

A section $D \in T_{\mathcal{S}}(U)$ is an “infinitesimal semantic variation” at U : a rule for perturbing the program’s observable behaviour at U to first order while obeying the Leibniz rule.

4.3 Tangent vectors at program coordinates

We describe $T_{\mathcal{S}}(U)$ concretely for the main coordinate types.

Example 4.3 (Function coordinate). Let U_f be the coordinate for a function $f: A \rightarrow B$ with specification $\varphi_f(x, y) \in \mathcal{O}_{\mathcal{S}}(U_f)$. A tangent vector $D \in T_{\mathcal{S}}(U_f)$ is an infinitesimal perturbation of the specification: $\varphi_f(x, y) \mapsto \varphi_f(x, y) + \varepsilon \cdot D(\varphi_f)(x, y)$, where $\varepsilon^2 = 0$. Examples include:

- Weakening a precondition: $D(\varphi_f) = -\delta P(x)$ for a predicate δP .
- Strengthening a postcondition: $D(\varphi_f) = +\delta Q(x, y)$.
- Renaming a specification variable: D acts as a substitution on specification identifiers.

Example 4.4 (Module coordinate). Let U_M be a module. The tangent space $T_{\mathcal{S}}(U_M)$ records *interface deformations*: infinitesimal changes to the exported type signatures and module contracts. Adding or removing a parameter from a public function contributes a non-trivial derivation.

Example 4.5 (Branch coordinate). For a conditional U_{if} with branches $U_{\text{then}}, U_{\text{else}}$, a tangent vector $D \in T_{\mathcal{S}}(U_{\text{if}})$ records an infinitesimal shift of the branch condition: a small perturbation of the guard predicate.

4.4 Restriction of tangent vectors

For a morphism $\iota: V \hookrightarrow U$ (inclusion of a subregion), the restriction map $\iota^*: T_{\mathcal{S}}(U) \rightarrow T_{\mathcal{S}}(V)$ sends $D \mapsto D|_V$ by composing with the restriction of $\mathcal{O}_{\mathcal{S}}$. This makes $T_{\mathcal{S}}$ into a sheaf on $\mathcal{S}$.

5 First-Order Refactorings and $H^1(\mathcal{S}, T_{\mathcal{S}})$

5.1 First-order deformations

Definition 5.1 (First-order deformation of P_0). A *first-order deformation* of P_0 is a sheaf $\mathcal{P}$ on $\mathcal{S}$ of $k[\varepsilon]/(\varepsilon^2)$ -algebras, flat over $k[\varepsilon]/(\varepsilon^2)$, together with an isomorphism $\mathcal{P} \otimes_{k[\varepsilon]/(\varepsilon^2)} k \xrightarrow{\sim} \mathcal{O}_{\mathcal{S}}$.

Concretely, at each coordinate U , a first-order deformation is an $\mathcal{O}_{\mathcal{S}}(U)[\varepsilon]/(\varepsilon^2)$ -module of the form

$$\mathcal{P}(U) = \mathcal{O}_{\mathcal{S}}(U)[\varepsilon]/(\varepsilon^2), \quad \text{with product } (f + g\varepsilon) \cdot (f' + g'\varepsilon) = ff' + (fg' + gf')\varepsilon.$$

The “direction” of the deformation at U is a derivation $D_U \in T_{\mathcal{S}}(U)$, and the deformed element $f + D_U(f)\varepsilon$ encodes the first-order change.

Example 5.2 (Method extraction). Let P_0 contain a function f whose body computes $e_1; e_2$. Extracting e_1 into a helper h is a first-order deformation: the derivation at U_f sends the body specification φ_f to $\varphi_f + \varepsilon \cdot \delta\varphi$, where $\delta\varphi$ records the introduction of the call to h .

5.2 Čech description of first-order deformations

Fix a cover $\mathcal{U} = \{U_i\}_{i \in I}$ of $\mathcal{S}$. A first-order deformation is specified by a Čech 1-cocycle:

Definition 5.3 (Čech 1-cocycle for $T_{\mathcal{S}}$). A *Čech 1-cocycle* with respect to $\mathcal{U}$ is a collection $\{g_{ij}\}_{i,j \in I}$ with $g_{ij} \in T_{\mathcal{S}}(U_i \cap U_j)$ satisfying the *cocycle condition*:

$$g_{ij} + g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k \quad \text{for all } i, j, k.$$

Two cocycles $\{g_{ij}\}$ and $\{g'_{ij}\}$ are *cohomologous* if there exist sections $\{h_i \in T_{\mathcal{S}}(U_i)\}_i$ with $g_{ij} - g'_{ij} = h_j|_{U_i \cap U_j} - h_i|_{U_i \cap U_j}$.

The equivalence classes form the group $H^1(\mathcal{S}, T_{\mathcal{S}})$, computed via the Čech complex $\check{C}^\bullet(\mathcal{U}, T_{\mathcal{S}})$:

$$0 \longrightarrow \check{C}^0(\mathcal{U}, T_{\mathcal{S}}) \xrightarrow{\delta^0} \check{C}^1(\mathcal{U}, T_{\mathcal{S}}) \xrightarrow{\delta^1} \check{C}^2(\mathcal{U}, T_{\mathcal{S}}) \longrightarrow \dots \quad (1)$$

where

$$(\delta^0 s)_{ij} = s_j|_{U_{ij}} - s_i|_{U_{ij}}, \quad (\delta^1 \varphi)_{ijk} = \varphi_{ij}|_{U_{ijk}} - \varphi_{ik}|_{U_{ijk}} + \varphi_{jk}|_{U_{ijk}},$$

and $U_{ij} = U_i \cap U_j$, $U_{ijk} = U_i \cap U_j \cap U_k$.

Proposition 5.4 (Chain complex identity). $\delta^1 \circ \delta^0 = 0$.

Proof. Direct computation: for $s \in \check{C}^0(\mathcal{U}, T_{\mathcal{S}})$,

$$(\delta^1(\delta^0 s))_{ijk} = (s_j - s_i) - (s_k - s_i) + (s_k - s_j) = 0. \quad \square$$

□

Theorem 5.5 (Classification of first-order refactorings). *There is a natural bijection between isomorphism classes of first-order deformations of P_0 and $H^1(\mathcal{S}, T_{\mathcal{S}})$.*

Proof sketch. The standard argument via Čech cohomology [?]: given a deformation, the gluing data on overlaps U_{ij} yields a 1-cocycle; two deformations are isomorphic iff their cocycles are cohomologous (differ by a coboundary). □

The *Kodaira-Spencer map* ρ_0 associates to a one-parameter family $\{P_t\}_t$ the class $[\partial_t|_{t=0}] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ of the derivative of the transition functions. We henceforth identify first-order refactorings with elements of $H^1(\mathcal{S}, T_{\mathcal{S}})$.

6 Obstructions in $H^2(\mathcal{S}, T_{\mathcal{S}})$

6.1 The extension problem

Given $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$, we ask:

Does $[\delta]$ extend to a second-order deformation over $k[t]/(t^3)$? And if so, to a full formal family over $k[[t]]$?

A second-order extension requires solving for a 1-cochain $\varphi^{(2)} \in \check{C}^1(\mathcal{U}, T_{\mathcal{S}})$ such that $\delta^1 \varphi^{(2)} = \text{obs}(\delta)$, where $\text{obs}(\delta) \in \check{C}^2(\mathcal{U}, T_{\mathcal{S}})$ is the *obstruction 2-cocycle*.

6.2 The obstruction cocycle

For a first-order deformation represented by a 1-cocycle $\{\varphi_{ij}\} \in \check{C}^1(\mathcal{U}, T_{\mathcal{S}})$, the obstruction to second-order extension is computed by the *cup product* (Lie bracket) of the deformation with itself:

Definition 6.1 (Obstruction cocycle). The obstruction 2-cochain is

$$\text{obs}(\varphi)_{ijk} = [\varphi_{ij}, \varphi_{jk}] \in T_{\mathcal{S}}(U_{ijk}),$$

where $[-, -]$ is the commutator of derivations in $T_{\mathcal{S}}(U_{ijk})$.

Lemma 6.2. $\text{obs}(\varphi)$ is a 2-cocycle: $\delta^2(\text{obs}(\varphi)) = 0$.

Proof. The cocycle condition for $\{\varphi_{ij}\}$ implies via the Jacobi identity for the Lie bracket that the alternating sum $\text{obs}(\varphi)_{jkl} - \text{obs}(\varphi)_{ikl} + \text{obs}(\varphi)_{ijl} - \text{obs}(\varphi)_{ijk} = 0$. $\square$

Definition 6.3 (Kodaira-Spencer class). The *Kodaira-Spencer obstruction class* of $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ is the cohomology class

$$\kappa([\delta]) = [\text{obs}(\varphi)] \in H^2(\mathcal{S}, T_{\mathcal{S}}),$$

where φ is any representative cocycle for $[\delta]$. The class is well-defined independently of the choice of representative.

6.3 Unobstructed programs

The second cohomology $H^2(\mathcal{S}, T_{\mathcal{S}})$ depends on the topology of the semantic site. For programs with simple cover structure, $H^2(\mathcal{S}, T_{\mathcal{S}})$ often vanishes.

Definition 6.4 (Unobstructed program). A program P_0 is *unobstructed* if $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$.

Proposition 6.5. *A program P_0 is unobstructed in the following cases:*

1. *The covering $\mathcal{U}$ has depth at most 2 (i.e., all triple overlaps U_{ijk} are empty).*
2. *The semantic site is acyclic: all higher cohomology of $T_{\mathcal{S}}$ vanishes.*
3. *Every function in P_0 has an independent specification (i.e., no shared semantic observables across three or more functions).*

Proof. In case (a), $\check{C}^2(\mathcal{U}, T_{\mathcal{S}}) = 0$ trivially. Cases (b) and (c) follow from the Leray spectral sequence for Čech cohomology: if the site has cohomological dimension at most 1, $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$. $\square$

7 The Kodaira-Spencer Theorem for Programs

We are now ready to state and prove the main result.

Theorem 7.1 (Kodaira-Spencer Criterion for Refactoring). *Let P_0 be a program on the semantic site $\mathcal{S}$, and let $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ be a first-order refactoring. The following are equivalent: (i)(i)*

1. *$[\delta]$ extends to a complete one-parameter refactoring family $\{P_t\}_{t \in [0,1]}$ with P_0 as the base fibre.*
2. *The Kodaira-Spencer class $\kappa([\delta]) \in H^2(\mathcal{S}, T_{\mathcal{S}})$ vanishes.*

Proof. (i) $\Rightarrow$ (ii): Given a full family $\{P_t\}$, the second-order transition data provides a 1-cochain $\varphi^{(2)}$ with $\delta^1 \varphi^{(2)} = \text{obs}(\varphi^{(1)})$. Hence $\text{obs}(\varphi^{(1)})$ is exact in $\check{C}^2(\mathcal{U}, T_{\mathcal{S}})$, so $\kappa([\delta]) = 0$.

(ii) $\Rightarrow$ (i): Suppose $\kappa([\delta]) = 0$. Then $\text{obs}(\varphi^{(1)}) = \delta^1 \varphi^{(2)}$ for some $\varphi^{(2)} \in \check{C}^1(\mathcal{U}, T_{\mathcal{S}})$. Set $\varphi^{(2)}$ as the second-order correction; the deformation equations are satisfied to order t^2 . Iterating this argument order by order, the obstruction at each stage lies in $H^2(\mathcal{S}, T_{\mathcal{S}})$; when it vanishes, we find corrections at every order and glue them via the formal power series completion $k[[t]]$ to obtain a full family. $\square$

Corollary 7.2 (Unobstructed programs). *If $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$, then every first-order refactoring $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ extends to a complete refactoring family.*

Proof. When $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$, every obstruction class is zero by definition. Apply Theorem ?? (ii) $\Rightarrow$ (i). $\square$

Corollary 7.3 (Composition of safe refactorings). *If $[\delta_1]$ and $[\delta_2]$ both have vanishing KS class, then $[\delta_1 + \delta_2]$ also has vanishing KS class provided that*

$$\kappa([\delta_1 + \delta_2]) = \kappa([\delta_1]) + \kappa([\delta_2]) + \text{cross terms},$$

and the cross terms vanish (e.g., when δ_1 and δ_2 act on disjoint coordinates).

The following diagram summarises the Kodaira-Spencer mechanism:

$$\begin{array}{ccc} H^1(\mathcal{S}, T_{\mathcal{S}}) & \xrightarrow{\kappa} & H^2(\mathcal{S}, T_{\mathcal{S}}) \\ & \searrow \text{extends?} & \uparrow \\ & & \{0\} \end{array}$$

A first-order refactoring extends if and only if its image under κ lands in the zero subgroup.

8 Application: Safe Refactoring

8.1 The safety condition

Recall from Paper 3 [?] that the verification status of P_0 is encoded as a section of the judgment sheaf $\mathcal{J}$ on $\mathcal{S}$. A refactoring *preserves verification* if the section $[\mathcal{J}]$ of the verification sheaf is unchanged modulo the deformation.

Definition 8.1 (Safe refactoring). A first-order refactoring $[\delta] \in H^1(\mathcal{S}, T_{\mathcal{S}})$ is *safe* if $\kappa([\delta]) = 0$ in $H^2(\mathcal{S}, T_{\mathcal{S}})$.

Theorem ?? guarantees that a safe refactoring extends to a full family without introducing verification failures at any order.

8.2 Algorithm

Algorithm ?? describes the pre-flight safety check.

Listing 1: Safe-refactoring pre-flight check.

```

1 def safe_refactoring_check(P0, refactoring) -> bool:
2     """
3     Pre-flight check: compute the Kodaira-Spencer obstruction.
4     Returns True iff the refactoring is guaranteed safe.
5     """
6     site = SemanticSite(P0)           # build S from P0
7     cover = site.compute_cover()      # covering family {U_i}
8     delta = encode_tangent(           # encode refactoring as
9         refactoring, site, cover)     # Čech 1-cocycle in H^1(S, T_S)
10
11     # Verify cocycle condition
12     if not is_cocycle(delta, cover):
13         raise ValueError("Refactoring does not satisfy cocycle
14                             condition")
15
16     # Compute the KS obstruction 2-cocycle obs = [delta, delta]
17     obs = compute_obstruction(delta, cover) # in Čech^2
18
19     # Check whether obs is exact: does delta^1(phi2) = obs have a
20     # solution?
21     ks_class = cech_cohomology_class(obs, cover, degree=2)
22
23     return ks_class.is_zero()         # True => safe, False => obstructed

```

8.3 Computing the Čech complex

For a program P_0 with n top-level functions, the covering $\mathcal{U}$ has:

- n objects (one per function),
- $\binom{n}{2}$ pairwise overlaps (shared module/interface dependencies),
- $\binom{n}{3}$ triple overlaps.

The Čech complex $\check{C}^\bullet(\mathcal{U}, T_S)$ has dimensions n , $\binom{n}{2}$, $\binom{n}{3}$ respectively. For typical programs where most functions are independent, most overlaps are empty and the complex is sparse. Computing $H^2(\mathcal{S}, T_S)$ reduces to a finite linear algebra problem over the integers.

8.4 Correctness

Theorem 8.2 (Safety certificate). *If $\text{safe_refactoring_check}(P_0, R)$ returns True, then the refactoring R applied to P_0 :*

1. *Preserves the verification status of every judgment $\mathcal{J} \in H^0(\mathcal{S}, \mathcal{J})$ verified before the refactoring.*
2. *Does not introduce new Čech obstructions in $H^1(\mathcal{S}, \mathcal{J})$ (from Paper 3 [?]).*
3. *Extends to a complete one-parameter family $\{P_t\}$ with P_0 as the base.*

Proof. Items (a)–(b) follow from Theorem ?? and the naturality of the map from deformations of $\mathcal{O}_S$ to deformations of $\mathcal{J}$. Item (c) is Theorem ?? directly. $\square$

8.5 Complexity

Let n be the number of coordinates in $\mathcal{S}$ and m the number of covering morphisms. Computing $H^2(\mathcal{S}, T_{\mathcal{S}})$ via the Čech complex requires:

- $O(m^2)$ to form $\check{C}^2(\mathcal{U}, T_{\mathcal{S}})$,
- $O(m^3)$ to compute $\ker(\delta^2)/\text{im}(\delta^1)$ by linear algebra (Gaussian elimination over the integers).

For sparse programs (i.e., $m = O(n)$), this is $O(n^3)$. In practice, the Gröbner-basis structure of the semantic algebra reduces the effective rank substantially.

9 Examples

9.1 Example 1: Method extraction (unobstructed)

Setting. P_0 contains a module M with two functions f and g sharing no interface dependencies. We wish to extract a helper h from the body of f .

Covering. $\mathcal{U} = \{U_f, U_g\}$. The overlap $U_f \cap U_g$ records only the module signature. The triple overlap $U_f \cap U_g \cap U_h$ (after introducing U_h) inherits only the module header.

Analysis. The tangent field δ at U_f is the derivation corresponding to replacing the inlined computation by a call to h . Since f and g share no deep interface:

$$\varphi_{fg} \in T_{\mathcal{S}}(U_f \cap U_g) \cong \{0\}, \quad \text{obs}(\varphi)_{fgh} = 0.$$

Hence $\kappa = 0$ and the extraction is safe.

Formal conclusion. By Corollary ??, since $H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$ in this case (depth-2 cover), the extraction extends globally.

9.2 Example 2: Interface refactoring (potentially obstructed)

Setting. Three functions f_1, f_2, f_3 all depend on a shared interface type τ . We refactor τ by adding a new field v .

Covering. $\mathcal{U} = \{U_{f_1}, U_{f_2}, U_{f_3}\}$. All three pairwise overlaps and the triple overlap are non-trivial: $T_{\mathcal{S}}(U_{ij}) \neq 0$ for all $i \neq j$, and $T_{\mathcal{S}}(U_{123}) \neq 0$.

Analysis. The derivation δ_{ij} at each overlap records the change to the interface type. The obstruction 2-cochain is $\text{obs}(\varphi)_{123} = [\varphi_{12}, \varphi_{23}]$, which in general is non-zero.

Condition for safety. The refactoring is safe iff the derivations on pairwise overlaps commute: $[\varphi_{12}, \varphi_{23}] = 0 \in T_{\mathcal{S}}(U_{123})$. This holds, for instance, when the interface changes are purely additive (no interaction between the new field v and existing fields).

9.3 Example 3: Loop invariant strengthening (unobstructed)

Setting. P_0 has a loop ℓ with invariant $I(x)$. We wish to strengthen the invariant to $I(x) \wedge J(x)$ where J is a new predicate that the loop body already preserves.

Analysis. The tangent vector $D \in T_{\mathcal{S}}(U_{\ell})$ records the addition of J to the invariant: $D(I) = J$. Since J is already preserved by the loop body, the deformation condition is satisfied with zero overlap data. The obstruction vanishes: $\kappa = 0$.

Formal conclusion. The strengthened invariant is a safe refactoring, and by Theorem ?? it extends to a full verification family.

10 Lean Formalization

All key algebraic results of this paper are formalized in LEAN 4 (Paper 22, `Paper22.DeformationTheory.lean`). The formalization uses the following model:

- `CochainFn0`, `CochainFn1`, `CochainFn2`: function types $\text{Coord} \rightarrow \text{TangentVal}$, $\text{Coord} \rightarrow \text{Coord} \rightarrow \text{TangentVal}$, $\text{Coord}^3 \rightarrow \text{TangentVal}$, where $\text{TangentVal} = \mathbb{Z}$.
- `, :` the coboundary operators.
- `chain_complex_identity`: $\delta^1 \circ \delta^0 = 0$, proved by `omega` (linear integer arithmetic).
- `coboundary_is_cocycle`: every coboundary is a cocycle.
- `FirstOrderDeformation`: a structure packaging a 1-cocycle.
- `DeformationProblem`: packages the obstruction 2-cocycle.
- `h2_zero_all_extend`: the main unobstructed theorem.

Table ?? summarizes the LEAN 4 theorems and their proof strategies.

Table 1: LEAN 4 formalization summary (0 sorry).

Theorem	Statement	Proof
<code>chain_complex_identity</code>	$\delta^1(\delta^0 s) = 0$	<code>omega</code>
<code>coboundary_is_cocycle</code>	$\text{im } \delta^0 \subseteq \ker \delta^1$	<code>simp+omega</code>
<code>delta0_antisymmetric</code>	$(\delta^0 s)_{ij} = -(\delta^0 s)_{ji}$	<code>omega</code>
<code>delta0_diagonal_zero</code>	$(\delta^0 s)_{ii} = 0$	<code>omega</code>
<code>deformation_equiv_refl</code>	$d \sim d$	explicit witness
<code>deformation_equiv_symm</code>	$d_1 \sim d_2 \Rightarrow d_2 \sim d_1$	negate section
<code>cocycle_add</code>	$\varphi, \psi \in \ker \delta^1 \Rightarrow \varphi + \psi \in \ker \delta^1$	<code>omega</code>
<code>trivial_is_extendable</code>	trivial deformation extends	zero witness
<code>h2_zero_all_extend</code>	$H^2 = 0 \Rightarrow$ all deformations extend	direct application
<code>unobstructed_implies_safe</code>	$H^2 = 0 \Rightarrow$ every refactoring is safe	from above
<code>sum_extendable</code>	sum of extendable problems is extendable	<code>omega</code>

11 Related Work

Deformation theory in algebra and geometry. Deformation theory originates with Kodaira-Spencer [?] for complex manifolds and Schlessinger [?] for algebraic functors. Grothendieck [?] generalized the theory to arbitrary ringed sites. Our adaptation to programs follows the Grothendieck approach: programs are sheaves on sites, and deformations are flat lifts.

Program transformation and verification. The semantics-preserving nature of refactoring is studied in the program transformation literature [?, ?]. These works characterize safe refactorings by preconditions on the program’s call graph or type structure. Our approach is different: we characterize safety via a cohomological invariant, which is *computable* and *global* (it accounts for all interactions, not just local ones).

Geometric approaches to program analysis. Several authors have proposed using topological or geometric notions for program analysis [?, ?]. Closest to our work is the persistent-homology approach of Curry [?], which studies sheaves on graphs. Our semantic site is more structured (Grothendieck topology rather than a simplicial complex) and our focus on the tangent sheaf is new.

Earlier Judgment Geometry papers. This paper builds directly on Papers 1 [?], 3 [?], and the broader verification framework of Papers 4–21. The safe-refactoring application connects deformation theory to the practical verification tools developed in Paper 10 [?].

12 Conclusion

We have developed a deformation-theoretic account of program refactoring. The key contributions are:

1. The tangent sheaf $T_{\mathcal{S}}$ of the semantic site, which classifies infinitesimal semantic variations.
2. The identification of first-order refactorings with $H^1(\mathcal{S}, T_{\mathcal{S}})$.
3. The Kodaira-Spencer obstruction $\kappa: H^1(\mathcal{S}, T_{\mathcal{S}}) \rightarrow H^2(\mathcal{S}, T_{\mathcal{S}})$, which is the complete obstruction to extension (Theorem ??).
4. The notion of *unobstructed programs* ($H^2(\mathcal{S}, T_{\mathcal{S}}) = 0$), for which every local refactoring extends globally (Corollary ??).
5. A pre-flight safety algorithm (Section ??) that certifies a refactoring before application.

The framework suggests several directions for future work: computing $H^2(\mathcal{S}, T_{\mathcal{S}})$ efficiently for large codebases; relating the Kodaira-Spencer class to traditional refactoring pre-conditions; and extending the theory to non-commutative semantic algebras (e.g., for concurrent programs).

References

- [1] K. Kodaira and D. C. Spencer. On deformations of complex analytic structures, I–II. *Annals of Mathematics*, 67(2):328–466, 1958.
- [2] M. Schlessinger. Functors of Artin rings. *Transactions of the American Mathematical Society*, 130(2):208–222, 1968.
- [3] R. Hartshorne. *Algebraic Geometry*. Springer-Verlag, New York, 1977. Graduate Texts in Mathematics 52.
- [4] A. Grothendieck. *Revêtements Étales et Groupe Fondamental* (SGA 1). Springer-Verlag, 1971. Lecture Notes in Mathematics 224.
- [5] W. G. Griswold and D. Notkin. Automated assistance for program restructuring. *ACM Transactions on Software Engineering and Methodology*, 2(3):228–269, 1993.
- [6] W. F. Opdyke. *Refactoring Object-Oriented Frameworks*. PhD thesis, University of Illinois at Urbana-Champaign, 1992.

- [7] R. Ghrist. *Elementary Applied Topology*. Createspace, 2014.
- [8] J. Curry. *Sheaves, Cosheaves and Applications*. PhD thesis, University of Pennsylvania, 2014.
- [9] H. Young. Grothendieck topologies for compositional program analysis. *Judgment Geometry*, Paper 1, 2024.
- [10] H. Young. Cohomological diagnostics: Classifying why proofs fail. *Judgment Geometry*, Paper 3, 2024.
- [11] H. Young. An empirical study of sheaf-theoretic program verification. *Judgment Geometry*, Paper 10, 2024.